Statistical Foundations of Machine Learning

Slow-Paced Tutorial: Session 1

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Concept Review

Expectation

For a discrete random variable X,

$$\mathbb{E}[X] = \sum_{x \in S_X} x p_X(x).$$

For a continuous random variable X,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

More generally,

$$\mathbb{E}[g(X)] = \sum_{x \in S_X} g(x) p_X(x)$$

in the discrete case, and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

in the continuous case.

Variance

If

$$\mu_X = \mathbb{E}[X],$$

then

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mu_X)^2\right].$$

The useful computational formula is

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - \left(\mathbb{E}[X]\right)^2.}$$

Also,

$$\boxed{\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X)}$$

If $X_1, \ldots, X_m$ are independent and have the same variance σ^2, then

$$\mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m} X_i\right) = \frac{\sigma^2}{m}.$$

Important Distributions

Distribution	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
$\mathrm{Ber}(p)$	p	$p(1-p)$
$\mathrm{Unif}(a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2
$\mathrm{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Random Vector

A random vector (X, Y): The components may or may not be independent.

Joint CDF

The joint cumulative distribution function is

$$F_{XY}(x, y) = P(X \le x,\ Y \le y).$$

The marginal CDFs are

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x, y), \qquad F_Y(y) = \lim_{x \to \infty} F_{XY}(x, y).$$

For $a < b$ and $c < d$,

$$\begin{aligned} &P(a < X \le b,\ c < Y \le d) \\ &= F_{XY}(b, d) - F_{XY}(a, d) \\ &\quad - F_{XY}(b, c) + F_{XY}(a, c). \end{aligned}$$

Discrete Random Vectors

For discrete X and Y, the joint PMF is

$$p_{XY}(x, y) = P(X = x,\ Y = y).$$

It satisfies

$$\sum_x \sum_y p_{XY}(x, y) = 1.$$

The marginal PMFs are obtained by summing:

$$p_X(x) = \sum_y p_{XY}(x, y),$$

$$p_Y(y) = \sum_x p_{XY}(x, y).$$

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Conditional PMF

If $p_Y(y) > 0$, then

$$p_{X|Y}(x \mid y) = \frac{p_{XY}(x,y)}{p_Y(y)}.$$

Joint pmf

marginal of Y

For every fixed y,

$$\sum_x p_{X|Y}(x \mid y) = 1.$$

The law of total probability gives

$$p_X(x) = \sum_y p_{X|Y}(x \mid y) p_Y(y).$$

Independence

The random variables X and Y are independent if

$$F_{XY}(x,y) = F_X(x) F_Y(y)$$

for every x, y.

For discrete random variables,

X and Y are independent $\iff$ $p_{XY}(x,y) = p_X(x) p_Y(y).$

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x), \qquad p_Y(y) > 0.$$

Jointly Continuous Variables

If (X, Y) has joint density f_{XY}, then

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy,$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dx.$$

If $f_Y(y) > 0$, the conditional density is

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}.$$

For jointly continuous random variables,

X and Y are independent $\iff$ $f_{XY}(x,y) = f_X(x) f_Y(y)$

almost everywhere.

$F(x) = P(X \le x)$ ← CDF

CDF $= \int_{-\infty}^{x} f_x(x)\, dx$

pdf

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Practice Questions

Q1. Let X be a discrete random variable with PMF

$$p_X(x) = \begin{cases} \dfrac{1}{n}, & x \in \{1, 2, \ldots, n\}, \\ 0, & \text{otherwise}, \end{cases}$$

where $n \geq 2$ is an integer. Find

$$\mathbb{E}[X] \qquad \text{and} \qquad \mathrm{Var}(X).$$

Q2. A fair coin is tossed independently three times. Let X denote the number of tails obtained, and define

$$Y = X^2, \qquad Z = 2X^2 + 1.$$

Find

$$\mathbb{E}[Y], \qquad \mathrm{Var}(Y), \qquad \mathbb{E}[Z], \qquad \mathrm{Var}(Z).$$

Q3. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{otherwise}. \end{cases}$$

(i) Verify that f_X is a valid PDF.

(ii) Calculate $\mathbb{E}[X]$.

(iii) Calculate $\mathbb{E}[X^2]$.

(iv) Hence find $\mathrm{Var}(X)$ and the standard deviation of X.

Q4. The joint PDF of (X, Y) is

$$f_{XY}(x, y) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1, \\ 0, & \text{otherwise}. \end{cases}$$

(i) Find the marginal PDFs f_X and f_Y.

(ii) Determine whether X and Y are independent.

(iii) Calculate

$$P\left(0 < X < \frac{1}{2}, \ \frac{1}{4} < Y < 1\right).$$

(iv) Calculate

$$P(X + Y < 1).$$

Q5. Let X and Y be random variables satisfying

$$P(X \in \{0, 1\}) = P(Y \in \{0, 1\}) = 1.$$

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Suppose that

$$P(X = 1, Y = 1) = P(X = 1)P(Y = 1).$$

Show that

$$p_{XY}(x, y) = p_X(x)p_Y(y)$$

for every $x, y \in \{0, 1\}$. Hence prove that X and Y are independent.

Q6. Let (X, Y) be a discrete random vector with joint PMF

$$p_{XY}(x, y) = \begin{cases} \dfrac{x + 2y}{18}, & x, y \in \{1, 2\}, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Verify that p_{XY} is a valid joint PMF.

(ii) Find the marginal PMFs p_X and p_Y.

(iii) Find the conditional PMFs

$$p_{Y|X}(y \mid 1) \qquad \text{and} \qquad p_{Y|X}(y \mid 2).$$

(iv) Determine whether X and Y are independent.

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Q1. Let X be a discrete random variable with PMF

$$p_X(x) = \begin{cases} \frac{1}{n}, & x \in \{1, 2, \ldots, n\}, \\ 0, & \text{otherwise}, \end{cases}$$

Support S_X

where $n \geq 1$ is an integer. Find

$$\mathbb{E}[X] \quad \text{and} \quad \text{Var}(X).$$

$$E[X] = \sum_{x \in S_X} x\, P_X(x) = \sum_{x \in S_X} x \frac{1}{n}$$

$$= \frac{1}{n} \boxed{\sum_{x \in S_X} x}$$

$$= \frac{1}{n} \sum_{i=1}^{n} i \qquad 1+2+\ldots+n = \frac{n(n+1)}{2}$$

$$= \frac{1}{n} \cdot \frac{n(n+1)}{2} = \boxed{\frac{n+1}{2}}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$g(x) = x^2 \qquad E(g(X)) = \sum_{x \in S_X} g(x)\, P_X(x)$$

$$= \sum_{x \in S_X} x^2 \frac{1}{n} = \frac{1}{n}\left(1^2 + 2^2 + \ldots n^2\right)$$

$$= \frac{1}{n} \frac{n(n+1)(2n+1)}{6}$$

$$E[X^2] = \frac{(n+1)(2n+1)}{6}$$

$$\text{Var}(X) = E[X^2] - (EX)^2 = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2$$

$$= \boxed{\frac{n^2-1}{12}}$$

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Q2. A fair coin is tossed independently three times. Let X denote the number of tails obtained, and define

$$Y = X^2, \qquad Z = 2X^2 + 1.$$

Find

$$\mathbb{E}[Y], \quad \mathrm{Var}(Y), \quad \mathbb{E}[Z], \quad \mathrm{Var}(Z).$$

X = # of tail obtained $X \sim \text{Bino}(n=3, p=\frac{1}{2})$

$$b(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$n=3, p=\frac{1}{2}$ $k \in \{0,1,2,3\} = S_X$

$a^b, a^c = a^{b+c}$

$$p_X(k) = \binom{3}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{3-k}$$

$$= \binom{3}{k} \left(\frac{1}{2}\right)^3 = \frac{1}{8}\binom{3}{k}$$

$$p_X(k) = \frac{1}{8}\binom{3}{k} : k = \{0,1,2,3\}$$

k	0	1	2	3
$p_X(k)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$X \leftarrow$ pmf

$g(X) \leftarrow Eg(x) = \sum g(x) \text{Pmf}(x)$

$$E[X] = E[X^2] = \sum_{k \in \{0,1,2,3\}} k^2 \cdot p_X(k)$$

$:= E(g(X))$

$g(X) = X^2$

$$= 0 \cdot p_X(0) + 1^2 p_X(1) + 2^2 p_X(2) + 3^2 p_X(3)$$

$$= 0 + \frac{3}{8} + 4 \cdot \frac{3}{8} + 9 \cdot \frac{1}{8}$$

$$= \frac{3+12+9}{8} = 3$$

$$E[Y] = 3$$

$$\mathrm{Var}[Y] = E[Y^2] - (E[X])^2$$

$Y^2 = (X^2)^2 = X^4$ $g(X) = X^4$

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$$E[Y^2] = E[X^4] = 0^4 \cdot \frac{1}{8} + 1^4 \cdot \frac{3}{8} + 2^4 \cdot \frac{3}{8} + 3^4 \cdot \frac{1}{8}$$

$$= \frac{33}{2}$$

So. $\quad Var(Y) = \frac{33}{2} - (3)^2 = \frac{33}{2} - 9 = \boxed{\frac{15}{2}}$

$Z = 2X^2 + 1$

$= 2Y + 1$

$$E[Z] = 2E[Y] + 1$$

$$= 2 \cdot 3 + 1$$

$$= 7$$

$$Var(Z) = Var(2Y+1)$$

$$= 2^2 \, Var(Y)$$

$$= 4 \cdot \frac{15}{2}$$

$$= \boxed{30}$$

$Y = aX + b$

$$E[Y] = \int_{-\infty}^{\infty} y \, P_Y(y) \, dy$$

$$= \int_{-\infty}^{\infty} (ax+b) \, P_Y(y) \, dy$$

$$= a \int_{-\infty}^{\infty} x \quad + \int$$

$$= a \, E[X] + b$$

$$\begin{cases} Y = aX + b \\ Var(Y) = a^2 \, Var(X) \end{cases}$$

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Q3. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 6x(1-x), & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases} \quad \geq 0$$

(i) Verify that f_X is a valid PDF.

(ii) Calculate $\mathbb{E}[X]$.

(iii) Calculate $\mathbb{E}[X^2]$.

(iv) Hence find $\mathrm{Var}(X)$ and the standard deviation of X.

i) $f_X(x) \geq 0$ and $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

Already satisfied

$$\int_{-\infty}^{\infty} f_X(x)\,dx = \int_0^1 6x(1-x)\,dx$$

$$= 6\int_0^1 (x - x^2)\,dx$$

$$= 6\left[\left(\frac{x^2}{2} - \frac{x^3}{3}\right)\right]_0^1 \qquad \left(\frac{1}{2} - \frac{1}{3} - 0\right)$$

$$= 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1$$

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx = \int_0^1 x\, 6x(1-x)\,dx$$

$$= 6\int_0^1 (x^2 - x^3)\,dx$$

$$= 6\left(\frac{x^3}{3} - \frac{x^4}{4}\right)_0^1 = 6\left(\frac{1}{3} - \frac{1}{4}\right)$$

$$= \frac{1}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\,dx = \int_0^1 x^2\, 6x(1-x)\,dx$$

$E[X^2]$ → function of X, $g(X) = X^2$

$$= \boxed{\frac{3}{10}}$$

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$$\text{Var}(X) = E\left[\overset{a}{X} - \overset{b}{E(X)}\right]^2$$

$$= E[X^2] - (EX)^2$$

$$= \frac{3}{16} - \left(\frac{1}{2}\right)^2 = \frac{3}{10} - \frac{1}{4} = \frac{1}{20}$$

Standar deviation of $X = \sqrt{\text{var}(X)}$

$$= \sqrt{\frac{1}{20}}$$

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Q4. The joint PDF of (X, Y) is

$$f_{XY}(x,y) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

valid

(i) Find the marginal PDFs f_X and f_Y.

(ii) Determine whether X and Y are independent.

(iii) Calculate

$$P\left(0 < X < \frac{1}{2}, \frac{1}{4} < Y < 1\right).$$

(iv) Calculate

$$P(X + Y < 1).$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dx$$

$$= \int_0^1 4xy\, dy \qquad = \int_0^1 4xy\, dx$$

$$= 2x \qquad = 2y$$

$$f_X(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases} \qquad f_Y(y) = \begin{cases} 2y & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\boxed{f_{XY}(x,y) = f_X(x)\, f_Y(y)}$$

$$f_X(x)\, f_Y(y) = (2x)(2y) : \begin{matrix} 0<x<1 \\ 0<y<1 \end{matrix}$$

$$= \begin{cases} 4xy & 0<x<1, 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

$$= f_{XY}(x,y)$$

$\Rightarrow$ X and Y are independent

for cdf
$F_{XY}(x,y) = F_X(x) F_Y(y)$

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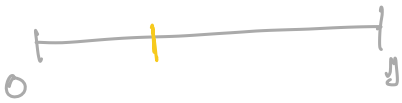
$$P\left(0<X<\tfrac{1}{2},\ \tfrac{1}{4}<Y<1\right)$$

$$P(X\in S_1,\ Y\in S_2) = P(X\in S_1)\,P(Y\in S_2)$$

$$= P\left(0<X<\tfrac{1}{2}\right)\,P\left(\tfrac{1}{4}<Y<1\right)$$

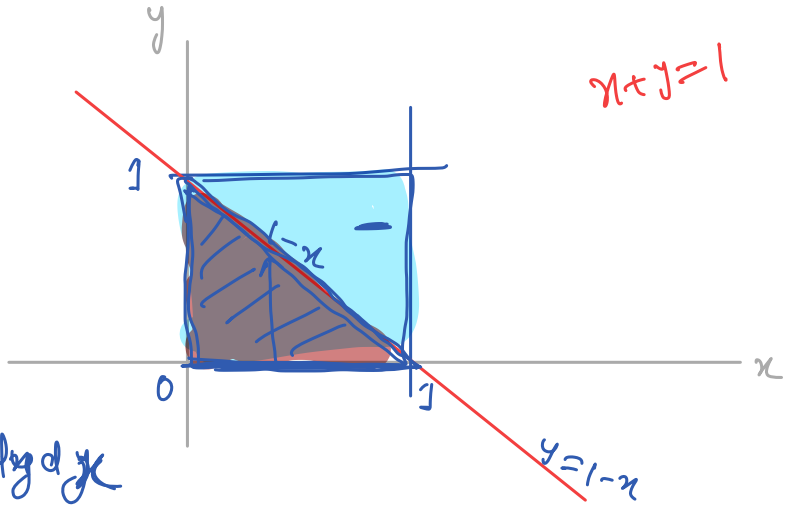
$$P\left(0<X<\tfrac{1}{2}\right) = \int_0^{1/2} 2x\,dx = \left[x^2\right]_0^{1/2} = \frac{1}{4}$$

$$P\left(\tfrac{1}{4}<Y<1\right) = \int_{1/4}^{1} 2y\,dy = \boxed{\frac{15}{16}}$$



$$P\left(0<X<\tfrac{1}{2},\ \tfrac{1}{4}<Y<1\right) = \frac{15}{16}\times\frac{1}{4} = \boxed{\frac{15}{64}}$$

$$\boxed{P(X+Y<1)}$$



$$= \int_{x=0}^{1}\int_{y=0}^{1-x} f_{XY}(x,y)\,dy\,dx$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} 4xy\,dy\,dx = \int_{x=0}^{1} 4x\left[\frac{y^2}{2}\right]_0^{1-x} dx$$

$$= \int_{x=0}^{1} 4x\,\frac{(1-x)^2}{2}\,dx = \boxed{\frac{1}{6}}$$

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Q5. Let X and Y be random variables satisfying

$$P(X \in \{0,1\}) = P(Y \in \{0,1\}) = 1.$$

Suppose that

$$P(X = 1, Y = 1) = P(X = 1)P(Y = 1).$$

Show that

$$p_{XY}(x, y) = p_X(x)p_Y(y)$$

for every $x, y \in \{0, 1\}$. Hence prove that X and Y are independent.

$P(X=1, Y=0)$ $P(X=0, Y=1)$ $P(X=0, Y=0)$

$=$ $=$ $=$

$P(X=1)\,P(Y=0)$

Let $a = P(X=1)$

$b = P(Y=1)$

Then $P(X=1, Y=1) = ab$

$$P(X=1, Y=0) = P(X=1)\,P(Y=0)$$

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Q6. Let (X, Y) be a discrete random vector with joint PMF

$$p_{XY}(x,y) = \begin{cases} \dfrac{x+2y}{18}, & x, y \in \{1,2\}, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Verify that p_{XY} is a valid joint PMF.

(ii) Find the marginal PMFs p_X and p_Y.

(iii) Find the conditional PMFs

$$p_{Y|X}(y \mid 1) \quad \text{and} \quad p_{Y|X}(y \mid 2).$$

(iv) Determine whether X and Y are independent.

$p_{X,Y}$ $p_{X|Y}$ p_{XY}

i) Need to check

a) $p_{XY} \geq 0$ ✓

b) $\sum_x \sum_y p_{XY}(x,y) = 1$

$p(x,y)$	X: 1	X: 2
Y: 1	$\frac{3}{18} = p_{XY}(1,1)$	$\frac{4}{18}$ $p_{XY}(1,2)$
Y: 2	$\frac{5}{18}$	$\frac{6}{18}$

≥ 0

$$\frac{3}{18} + \frac{4}{18} + \frac{5}{18} + \frac{6}{18} = \frac{18}{18} = 1$$

ii) Marginal

$$p_X(x) = \sum_{y \in \{1,2\}} p_{XY}(x,y) = \sum_{y \in \{1,2\}} \frac{x+2y}{18}$$

$$= \frac{x+2}{18} + \frac{x+4}{18} = \frac{2x+6}{18}$$

$$= \frac{x+3}{9}$$

$$\boxed{p_X(x) = \frac{x+3}{9} \quad x \in \{1,2\}}$$

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$$p_Y(y) = \sum_{x \in \{1,2\}} \frac{x+2y}{18} = \frac{1+2y}{18} + \frac{2+2y}{18}$$

$$= \frac{3+4y}{18}$$

$$\boxed{p_Y(y) = \frac{3+4y}{18} \quad y \in \{1,2\}}$$

$$p_{Y|X}(y|x) = \frac{p_{XY}(x,y)}{p_X(x)}$$

$$p_{Y|X}(y|1) = \frac{p_{XY}(y,1)}{p_X(1)}$$

$$= \frac{\frac{1+2y}{18}}{4/9} = \frac{1+2y}{8}$$

$$\boxed{p_{Y|X}(y|1) = \frac{1+2y}{8} \quad y \in \{1,2\}}$$

$p_{Y|X}(y|2)$ → similarly just as above

X and Y independent or not

$$p_{XY}(x,y) = p_X(x)\, p_Y(y)$$

$$p_{XY}(1,1) = \frac{1}{6} \qquad p_X(1) = \frac{4}{9}$$

$$p_Y(1) = \frac{7}{18}$$

$$\frac{1}{6} \neq \frac{4}{9} \cdot \frac{7}{18}$$

X and Y is not Independent

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Bivariate Gaussian

$X \sim N(0,1)$

$Y \sim N(0,1)$

ρ = correlation between X and Y

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\left[\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right]}$$

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